

Capstone Proposal — Orallexa Calibration

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Formal Capstone Proposal — Summer 2026

Title: From Paper to Production: Calibration Discipline in a Solo-Operated AI Trading Agent

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1. Problem statement

Most academic work on AI-driven trading evaluates models on backtests over historical data. The literature reports calibration metrics (Brier score, log-loss, expected calibration error) on closed datasets where the answer key is known. **What is rarely measured is whether these calibration claims survive the wild** — adversarial real-time markets where information staleness, latency, and other agents trading the same signal degrade predictive quality.

This capstone asks: *can a single-developer AI trading agent maintain measurable Brier-score improvement over a market-implied baseline when deployed in production with real capital at small sizes (\$300-\$5k), across two heterogeneous asset classes?*

2. Hypothesis

A multi-source agent pipeline — Claude Sonnet 4.6 probability extraction + technical indicators (ATR-based volatility floors) + Brier-feedback loops — will achieve a Brier score reduction of ≥ 0.05 versus a market-implied baseline (e.g. Polymarket midpoint, S&P 500 sector benchmark) over a 90-day live evaluation window, *if and only if* the agent’s paper P&L tracks backtest P&L for at least 14 days prior to real-money deployment.

The “if and only if” clause encodes the methodological discipline: backtest-paper divergence is the falsification criterion. If backtest claims do not reproduce in live paper, no real money is deployed.

3. Methodology

3.1 Two test surfaces

Surface	Asset class	Signal source	Brier baseline
Polymarket	Binary prediction markets	LLM probability + persistence filter + 10pp mispricing gate	Polymarket implied midpoint
SpaceX equity sector	8 pure-play tickers (RCLB, ASTS, LUNR, BKSJ, PL, RDW, LMT, LIN)	LLM directional + sentiment + ATR stops	Buy-and-hold SPCE basket

3.2 Calibration measurement

For each prediction p over outcome o in $\{0, 1\}$, the Brier score component is $(p - o)^2$. The agent’s daily Brier is averaged across resolved decisions; the baseline’s Brier is computed identically on the same decision space. **The deliverable metric is `Brier_agent` - `Brier_baseline`** over rolling 30-day windows.

3.3 Public pre-registration

Every prediction is timestamped on a public GitHub repository ([spacex-ipo-tracker](#)) before the event resolves. This eliminates the cherry-picking attack vector that contaminates retail quant blogs.

3.4 Risk controls (hard-coded)

- **3-loss kill switch:** 3 consecutive losing trades pauses live execution
- **\$300 cap on Polymarket:** total real-money exposure
- **No leverage:** spot-only on equity, no derivatives
- **ATR stop-loss floor:** minimum 2%, cap 15%, scales with 14-day Average True Range
- **Sector exposure cap:** no single sector >30% of bankroll

4. Deliverables (12 weeks)

1. **Weeks 1-2:** Complete paper-validation phase. Daily Brier weekly reports.
2. **Weeks 3-4:** Deploy first real-money position (\$20-\$50 size) on highest-confidence Polymarket signal. Begin live Brier audit.
3. **Weeks 5-8:** If paper tracks backtest, scale to \$1k. Add second asset surface (equity) with same calibration discipline.
4. **Weeks 9-10:** Methodology paper draft. Open-source v1.0 release.
5. **Weeks 11-12:** Final paper revision. Defense + portfolio documentation.

5. Expected outcomes (success + failure)

- **Positive result:** Brier improvement ≥ 0.05 across both surfaces $\rightarrow$ methodology paper publishable at student venues (ICML student workshop, NeurIPS spotlights).
- **Null result:** Brier improvement < 0.05 but Brier $\leq$ market baseline $\rightarrow$ methodology paper still publishable as “honest negative” case study.
- **Negative result:** Brier $>$ market baseline $\rightarrow$ post-mortem paper on *which* calibration assumption failed (instructive for future agent research).

6. Required resources

- **Computational:** existing personal MacBook + Anthropic API budget (~\$50/month, self-funded)
- **Data:** free APIs (yfinance, polymarket gamma-api, GitHub stars-history, HN Algolia, Reddit JSON)
- **Capital at risk:** \$300 (self-funded, hard-capped)
- **Advisor time:** estimated 1-2 hours per week for paper discussion + methodology critique

7. Why this works at a student budget

Foundation model APIs (Claude/GPT) make the marginal cost of probability extraction near-zero. The research bottleneck is no longer compute or data — it is **calibration discipline**: rigorous tracking of whether claimed accuracy survives production. That is a 1-person problem.

8. Related work (selected)

See `references/references.bib`.

Key prior work: Brier (1950) for the calibration metric foundation; Tetlock (2005) for expert calibration methodology; Hanson (2003) for prediction-market theory; Shinn et al. (2023) for Reflexion-style agent self-improvement loops; Sakana AI (2025) for Darwin-Gödel Machine self-modifying agents.